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Tip-dating and the origin of Telluraves

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ABSTRACT

Despite a relatively vast accumulation of molecular data, the timing of diversification of modern bird lineages remains elusive. Accurate dating of the origination of Telluraves—a clade of birds defined by their arboreality—is of particular interest, as it contains the most species-rich avian group, the passerines. Historically, neontological studies have estimated a Cretaceous origin for the group, but more recent studies have recovered Cenozoic dates, closer to the oldest known fossils for the group. We employ total-evidence dating to estimate divergence times that are expected to be both less sensitive to prior assumptions and more accurate. Specifically, we use a large collection of morphological character data from arboreal bird fossils, along with combined molecular sequence and morphological character data from extant taxa. Our analyses recover a Late Cretaceous origin for crown Telluraves, with a few lineages crossing the K-Pg boundary. Following the K-Pg boundary, our results show the group underwent rapid diversification, likely benefiting from increased ecological opportunities in the aftermath of the extinction event. We find very little confidence for the precise topological placement of many extinct taxa, possibly due to rapid diversification, paucity of character data, and rapid morphological differentiation during the early history of the group.

1. Introduction

Understanding how biodiversity has changed over time requires reliable estimates of divergence times between species. Molecular studies concerned with estimating clade ages most commonly do so by reconstructing phylogenetic trees which incorporate a model of molecular evolution that is calibrated by specified ages for nodes within the tree. This method is known as the 'node-dating' approach to phylogenetics and, although it is widely used, the manner in which extinct taxa are assigned to nodes can be problematic, and potentially distort results (Near et al., 2005; Arcila et al., 2015). Moreover, this approach ignores any extinct species that is not unambiguously identified as the oldest of its clade, as well as information on the morphology of extinct taxa. Alternatively, "total-evidence" approaches allow fossils to be placed as terminal taxa in a dating analysis using both molecular and morphological data (Pyron, 2011; Ronquist et al., 2012a). Doing so can lead to improved date estimates, particularly by breaking up longer branches (Heath et al., 2008; Ronquist et al., 2012a; Crawley and Hilu, 2012; Wright and Hillis, 2014). As different tip-dating approaches may yield both contrasting topological results and significantly older divergence estimates, careful examination of results remains important. (Arcila et al., 2015; Bapst et al., 2016).

Total-evidence represents a powerful tool for investigating evolutionary processes across the tree of life, including birds – the most diverse clade of terrestrial vertebrates. However, recent attempts to assess timing of divergence events across large avian clades have relied on node-dating methods and included very few extinct taxa relative to number of tips. For example, Jarvis et al. (2014) and Prum et al. (2015) included 39 and 19 node calibrations, respectively. Although more than previous studies, they do not reflect extinct biodiversity (Mayr, 2009), as they only sample the oldest members of extant clades. The absence of a large tip-dating study for crown species is principally due to the absence of a morphological matrix covering both the extent of the avian tree and the diversity of forms seen in the fossil record. Although construction of such matrices remain a time-consuming and challenging task (Scotland et al., 2003; Prum et al., 2015), their development offers potential for improving our understanding of these enigmatic taxa.

In this study, we construct a morphological matrix spanning the most speciose clade of the avian tree that also incorporates the majority of its described extinct taxa. We focus on is Telluraves ('earth birds'; Yuri et al., 2013; Jarvis et al., 2014). Specifically, we include taxa from eight orders: woodpeckers & allies (Piciformes), kingfishers & allies (Coraciiformes), hornbills & hoopoes (Bucerotiformes), trogons (Trogoniformes), cuckoo roller (Leptosomiformes), mousebirds

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(Coliiformes), parrots (Psittaciformes), and the passerines (Passeriformes). These orders represent a nearly complete spectrum of diversity within Telluraves, with only birds of prey not sampled (e.g. owls, Strigiformes, and falcons, Falconiformes). A rich and diverse collection of fossil Telluraves specimens have been cataloged over the last several decades (Mayr, 2009), allowing great insight into the timing of diversification within the group. We compile an extensive morphological data matrix, which contains 122 extinct taxa, including those described from nearly complete skeletons. The increasing availability of nucleic sequence data further enables us to include a great number of taxa in a combined nucleic sequence and morphological data matrix. Using this matrix, we estimate key dates in the diversification of Telluraves. Next, we examine the effect of the inclusion of extinct taxa on the inference of the topology and age estimates for Telluraves. Finally, we assess confidence in the placement of the extinct taxa within the tree, and compare our results to those of previous studies.

2. Materials and methods

2.1. Morphological character data

The morphological component of the data matrix was assembled using a combination of nine previously published character matrices as well as primary literature descriptions of fossil material. Previously published character matrices were from Clarke et al. (2009), Ksepka and Clarke (2010), DeBee (2012), Worthy et al. (2010), Mayr (2004, 2013), Mayr et al. (2004, 2013), Claramunt and Rinderknecht (2005). The data from Mayr et al. (2013) is an expanded and modified version of the data compiled by Ksepka and Clarke (2012). The collected matrices were concatenated so that homologous characters between studies were not repeated within the matrix. For example, character 34 (development of the procoracoid process) is shared between seven studies (supplementary material). Combining the scorings from across studies ensures that this trait does not contribute disproportionately to the results. If a species had been scored in multiple studies with different character states, we used the most recently assigned character state. We retained the 'ordered' coding for forty-four characters (enforcing transitions through intermediate states) from the published character matrices, which were coded as 'ordered' in the original studies. We also added 46 new characters which we defined based on primary literature descriptions of extinct taxa. Morphological characters were almost exclusively osteological, primarily from the skull, wing, and leg bones (Fig. 1). The matrix also contains fifteen soft-tissue and life history characters. We recorded data for all extinct species that we found, including recently extinct species, such as St. Helena hoopoe, *Upupa antaios* (Table S6). The final morphological component of the matrix contains recorded states for a total of 782 characters. For all characters, there was a mean of 2.12 states, and a mean of 3.23 states for ordered characters.

2.2. Molecular sequence data

We collected previously published DNA sequences, available on GenBank, and aligned them using a custom pipeline. The final alignment comprised of seven mitochondrial (*NADH2, ATP8, CytB, ATP6, NADH3, CO1, 16S*) and five nuclear loci (*C-mos, EGR1, RAG1, RAG2, TGBF2*), with a total alignment length of 27,033 base pairs (bp; Table S1). All sequences available for the chosen loci, for all species within Telluraves, were downloaded from GenBank. We selected the extant taxa using the following heuristic scheme aimed at maximizing coverage and computational efficiency. The minimum limit for inclusion was the presence of six loci in the alignment matrix, or half of the total alignment length. We then randomly subsampled the alignment matrix to retain the number of taxa in proportion to the size of the order within which they are found (Fig. S1). We made some exceptions. Taxa with fewer than six loci were retained if they were scored for a large number of morphological characters to help maximize overlap between the genetic and morphological data sets, facilitating placement of extinct taxa (Guillerme and Cooper, 2016). We also included five species of Caprimulgiformes (nightjars and allies) as an outgroup. The group contains species scored for a large number of morphological characters, and it is frequently used as an outgroup in phylogenetic studies of Telluraves (Mayr, 2004; Mayr et al., 2004; Clarke et al., 2009; DeBee, 2012). The complete list of loci used for each species is provided in Table S2.

We created consensus sequences for species with multiple sequences reported for a given locus. The complete set of sequences was first aligned using MAFFT v7.164 (Katoh et al., 2002). We used the generalized affine gap cost procedure with a gap opening penalty of 3 and a JTT 1-PAM number matrix (Jones et al., 1992). The procedure was repeated for 1000 cycles of iterative refinement. Consensus sequences were then created using the resulting alignments, which scored positions as ambiguous if they were represented by a single nucleotide in less than 60% of the sequences. Each species was represented by a single sequence for each locus in the alignment. We created alignments in this manner for each of the twelve loci individually, using consensus sequences as necessary. Locus alignments used the generalized affine gap cost method, with a BLOSUM-62 number matrix (Henikoff and Henikoff, 1992). The alignment of loci was also repeated for 1000 cycles of iterative refinement. The twelve individual locus alignments were subsequently concatenated to form a single alignment for all species in the analysis.

We performed an initial phylogenetic reconstruction using RAxML (Stamatakis, 2014) and visually inspected the output to check for spuriously located taxa. A species was determined to be incorrectly placed if it was not clustered within its commonly accepted family from the International Ornithological Congress (IOC) checklist (Gill and Donsker, 2013). We chose family as the taxonomic cutoff as the species included in this study have unambiguous assignment at this level. We identified nine species out of the 267 initially included extant taxa that appeared to be erroneously placed within the tree, and which we suspected to possibly result in "rogue" placement (e.g., Sanderson and Shaffer, 2002). Four of these species contained only a single gene in the DNA matrix (*Malacoptila fusca*, *Jamacerops aurea*, *Chelidoptera tenebrosa*, *Phoeniculus damarensis*), and two species contained only two genes (*Nystalus maculatus* and *Galbula cyanescens*). These six species were included in the preliminary analysis because they have a large number of scored morphological characters, but they were removed prior to the main analysis. These taxa are closely related, with other representatives of their families retained within the alignment of DNA sequence data (*Malacoptila semicincta*, *Galbula ruficauda*, *Phoeniculus purpureus*). The remaining three taxa were absent from the morphological alignment (*Psittaculirostris demarestii*, *Xipholena punicea* & *Rupicola peruviana*). There are a many possible causes for the incorrect positioning of the removed taxa, including incorrect labeling of sequences on GenBank, uncertain alignment, and gene tree incongruence. Due to the difficulty in identifying the source of the error, we used a heuristic approach, removing these three taxa prior to the main analysis.

2.3. Phylogenetic reconstruction and divergence time estimation

We implemented a relaxed clock model in MrBayes v3.2.3 (Ronquist et al., 2012b) for all phylogenetic analyses. Specifically, we used the Independent Gamma Rates model with an exponential prior for the variance parameter (using the default value of 10). We specified a log-normal prior on the substitution rate with a mean of -7.386 and a standard deviation of 1.000 following Ericson et al. (2014). For branch lengths, we assigned a uniform prior probability distribution.

We specified locus partitions for molecular sequence data within the concatenated alignment, with each partition assigned a GTR + I + Γ substitution model. For the morphological component of the matrix, we specified partitions based on main body regions: skull, body, wings,

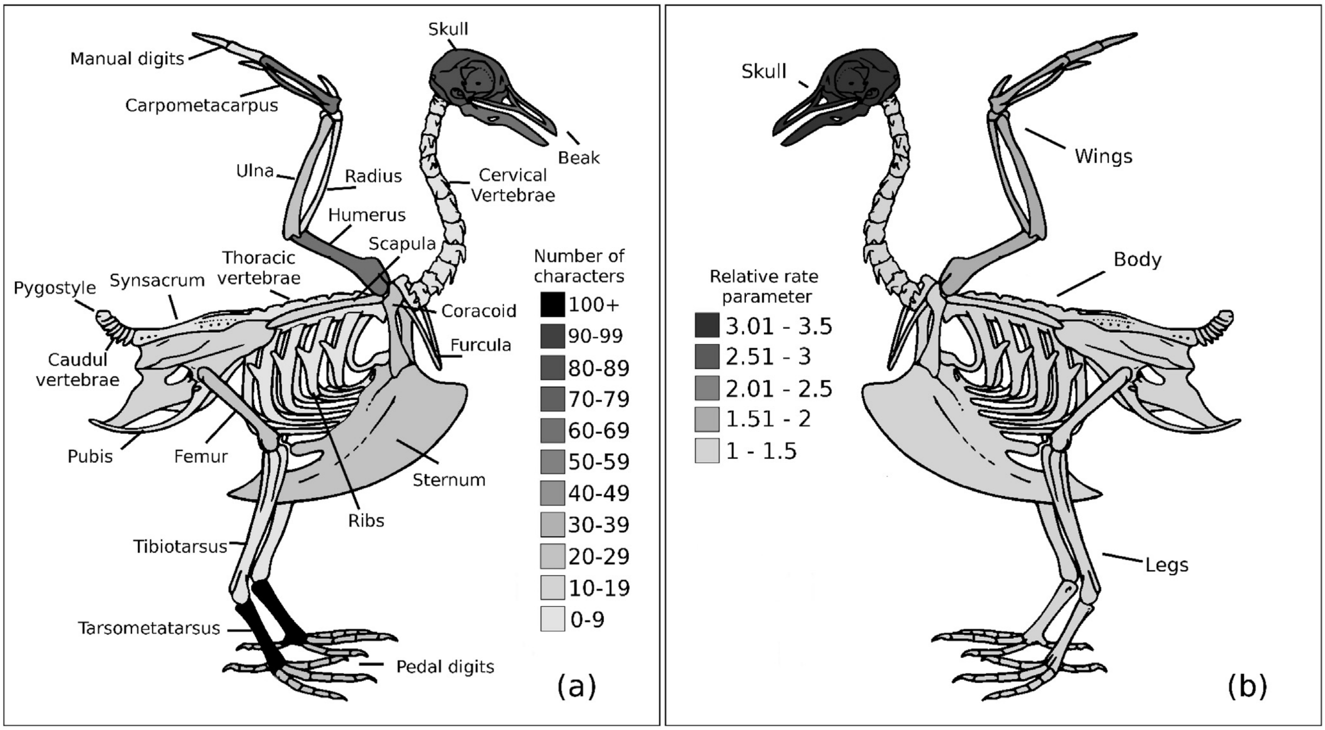

Fig. 1. Body region distribution of 767 scored osteological characters and relative rate parameters for four defined body region partitions. (a) The illustrated figures are shaded darker with the increasing number of scored characters from each feature. The tarsometatarsus had the highest number of characters in the morphological matrix (n = 198, around 25% of the total). Skull was scored for 156 characters, wings for 189, body for 115, and legs for 307. (b) The mean relative rate parameters for osteological characters, estimated under a MK + Γ substitution model, were roughly three times higher for the skull partition compared to wing, body, and leg characters (see Table 2 for addit.ional details).

legs, and 'other.' The 'other' partition included soft-tissue and life history characters. The partitions within the morphological matrix were coded as 'variable' due to the general lack of invariant characters present, a common characteristic of morphological data, compared to many molecular data for phylogenetic reconstruction (Lewis, 2001). We assigned a MK + Γ substitution model to each partition in the morphological matrix. The genetic and morphological partitions were unlinked, allowing independent parameter estimation for each partition. We placed uniform age priors on all of the extinct species. Uniform age priors follow Slater (2013), with fossil age estimates obtained from dates published on the Paleobiology Database (Behrensmeyer and Turner, 2013) or directly from the primary literature (Table S7). Generally, uniform priors allow us to accommodate a view that little is known about the time of divergence relative to the constraints. This as an untested hypothesis – that given the absence of evidence to the contrary, there is an equal prior probability for the timing of divergence, spanning the minimum and maximum age limits. This is not the unique case in which there is no material basis for selecting among the probability distributions and their parameters and, indeed, studies rarely or never can have truly justified priors (Warnock et al., 2012, 2014).

We did not enforce any topological constraints for the phylogenetic analysis. Topological constraints typically involve enforcing taxa to be monophyletic. Such constraints could be considered possible here given that there is little ambiguity as to the taxonomy of the taxa we have included. However, we left the placement of all taxa to be freely estimated by MrBayes so as to not potentially affect the estimated rate of evolution.

We first ran short runs of one million generations each to examine mixing of Markov chain Monte Carlo (MCMC) samples. We targeted acceptance rates of attempted swaps between the cold and heated chains between 20% and 80%, which achieved by adjusting the temperature parameter for the heated chains. The analysis was then run for 230 million generations using the following parameters: frequency of sampling the Markov chain (samplefreq) 1000, temperature for heating the chains (temp) 0.01, number of generations between check-pointing (checkfreq) 10,000, number of runs (nruns) 2, number of chains per run (nchain) 8, number of generations between the calculation of MCMC diagnostics (Diagnfreq) 2000, number of swaps tried for each swapping generation of the chain (Nswaps) 2, frequency of attempting swaps between chains (Swapfreq) 50. We estimated convergence on a stable posterior distribution by creating plots of the slowest converging parameters and seeing when they approached a stable distribution. We ensured that the minimum effective sample size for each parameter was greater than 100. To assess convergence in tree topology, we used standard deviation of split differences in topologies between different runs of the analysis, and an R version of 'Are We There Yet,' or 'rwty' (Nylander et al., 2008; Warren et al., 2015; R Core Team, 2013). Once the MCMC run reached a stable posterior distribution, we discarded the first 50% of steps as 'burn-in' and used the retained steps to calculate the summary statistics in MrBayes and 'rwty'. In addition to the analysis of the combined data set, we performed individual phylogenetic analyses on the DNA alignment matrix and combined morphological character state matrix (see Supplementary Materials).

2.4. Phylogenetic signal of morphological characters

As a broad estimate of phylogenetic signal, we calculated the λ metric Pagel (1999), an imperfect index of character lability of morphological characters (Revell et al., 2008). The value of λ gives a measure of similarity of observed covariances among species compared with the expected covariances under Brownian motion. It ranges between 0 (none) and 1 (identical to Brownian expectation). The likelihood of a model with λ freely estimated was compared against one in which the value of λ was constrained to 0.001, a model of vanishingly low phylogenetic signal, using a likelihood ratio test (Sundue and

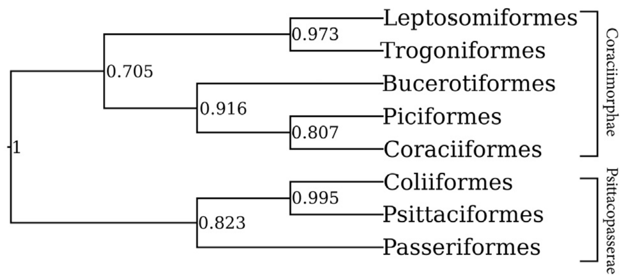

Fig. 2. Maximum clade credibility tree at the ordinal level from tip-dating analysis of the full dataset. Analysis of genetic data only results in Coliiformes being placed within the Coraciimorphae.

Rothfels, 2013). Many independent processes can generate a range of values of λ (Revell et al., 2008), and we merely employ it as a rough approximation of trait conservation.

Phylogenetic trees for these analyses were drawn from phylogenetic inference performed without morphological data, using only the molecular sequence alignment. These analyses were performed using the R package *geiger* (Harmon et al., 2008). We excluded invariable and ordinal characters for this analysis, as well as those characters with high tip imbalance – where > 90% of the species were scored in the same state. Finally, we removed any characters that were scored in fewer than 30 species to maximize the power of the analyses (Freckleton, 2002). We omitted 669 of 782 characters in this manner, leaving 113 in this analysis.

2.5. Placement of extinct taxa

To evaluate topological placement support for each extinct taxon, we calculated the frequency of its grouping within a taxonomic order by finding the nearest extant species to each extinct taxon, and recording the order (taxonomic rank) in which that extant species is classified. This procedure was repeated across all trees in the posterior phylogenetic set to calculate the frequency with which each extinct taxon is affiliated with each extant order, including the outgroup. The frequency with which each fossil is associated with its most commonly recorded order in the posterior sample of phylogenies is the placement support score it receives (Fig. S6).

To examine variation in placement support we included species values as the dependent variable in a one-way analysis of covariance (Wilkinson and Rogers, 1973, ANCOVA), implemented in R (Hector, 2015). We included the following independent variables: the superorder in which each species is found (Psittacopasserae or Coraciimorphae, Fig. 3), the number of characters scored, the minimum age of the species, the number of scored characters with $\lambda \geqslant 0.8$, as well as the diversity of body regions represented by the characters scored for a species. We determined the diversity of scored body regions for each species by counting the number of characters scored in each of the five body regions, and by calculating Simpson's diversity index in the R package *vegan* (Oksanen et al., 2016). Species received higher scores if the characters scored were distributed evenly across each of the five body regions.

3. Results

Out of the 782 characters in the morphological matrix, 721 characters (92.2%) were variable. The matrix coverage is fairly sparse, with 94.8% of cells missing data. This is, however, not unusual for palaeontological studies (Fig. S3), with the amount of missing data in this study following a general trend of matrix completeness seen in published studies (Fig. S3). There was high variation in the number of characters scored per species (Fig. S2), although those species with the fewest number of characters not ubiquitously extinct taxa. The mean number of scored characters per extinct species was 26, and the mean number of scored characters for extant species was 40. Species with large numbers of scored morphological characters were distributed evenly throughout the tree, with the largest numbers of extant taxa with no scored characters being predominantly Passeriformes and Psittaciformes (Fig. S4). Extant species had a mean of 5 loci in the genetic matrix (Fig. S5, Table S2), with 32% of the cells in the genetic matrix coded as missing. Species with fewer loci were frequently those which had been scored for a large number of morphological characters. Overlap between the morphological and genetic matrices was high, with 220 out of 258 extant species present in the morphological matrix.

Raw phylogenetic data output from MrBayes does not have branch lengths scaled to absolute time (in millions of years). We scaled the branch lengths using the estimated rate of evolution, with the procedure carried out independently for each sample in the posterior distribution as topology and rates of evolution are estimated simultaneously. We discarded all trees in the first 190 million generations, i.e. those samples prior to convergence. Therefore, after pooling the two runs together, and with an analysis sampling frequency of 1000, we retained 80,000 trees. From this posterior sample, we randomly subsampled 5000 trees for analyses.

3.1. Phylogenetic relationships

Inclusion of extinct taxa resulted in Coliiformes being recovered as sister to Psitaciiformes, with strong support (Fig. 2), and not within the Coraciimorphae (Hackett et al., 2008; Jarvis et al., 2014; Prum et al., 2015). Additionally, the topology estimated here recovers Trogoniformes and Leptosomiformes as sister taxa, but their close relationship is weakly supported (Fig. 2). Analysis of only the genetic data also estimates a sister relationship between these orders, however the Coliiformes are estimated to fall within the Coraciimorphae. This analysis also estimates Coraciiformes as sister to a clade of Bucerotiformes and Piciformes, rather than Bucerotiformes as sister to a clade of Coraciiformes and Piciformes (Fig. S7). Analysis of morphological data only did not produce significant support for the backbone topology.

3.2. Divergence estimates

Inclusion of extinct taxa dramatically influenced the effective prior estimated by MrBayes (calculated by running the mcmc chain with 'data = no' specified). Compared to an alignment of only genetic data the effective clock rate prior increased from 0.002 to 0.003, changing the effective tree height prior from 79.5 Ma to 125.7 Ma. Following analysis, we find a Telluraves crown mean age estimate of 75.6 Mya (95% HPD: 69.9–82.8 Mya), falling entirely within the Cretaceous (Table 1). The inferred crown ages of some orders also pre-date the K-Pg boundary, five of which have mean divergence estimates in the Cretaceous (Table 1). Only two orders (Piciformes & the monotypic Leptosomiformes) having the lower credible interval for their crown age preceding the K-Pg boundary by more than two million years. Another date of interest, the crown age for oscine and suboscine passerine species, spans the K-Pg boundary. The deepest nodes in the tree, those subtending the modern taxonomic orders, have short internode distances, perhaps indicative of high uncertainty or rapid early diversification for Telluraves (Jarvis et al., 2014; Prum et al., 2015). The estimated age gap between the oldest fossils and the root of our tree is approximately 20 My. The root age in analysis of morphological data only produced a root age similar to that of the full data set (mean 71.0 Mya, 66.4–77.0 95% HPD). Analysis using only the genetic data resulted in a somewhat younger, post-K-Pg estimate (mean 61.0 Mya, 35.0–88.7 95% HPD). BAMM diversification rate estimates based on the analysis including 122 extinct species support a post-K-Pg radiation (Fig. 4).

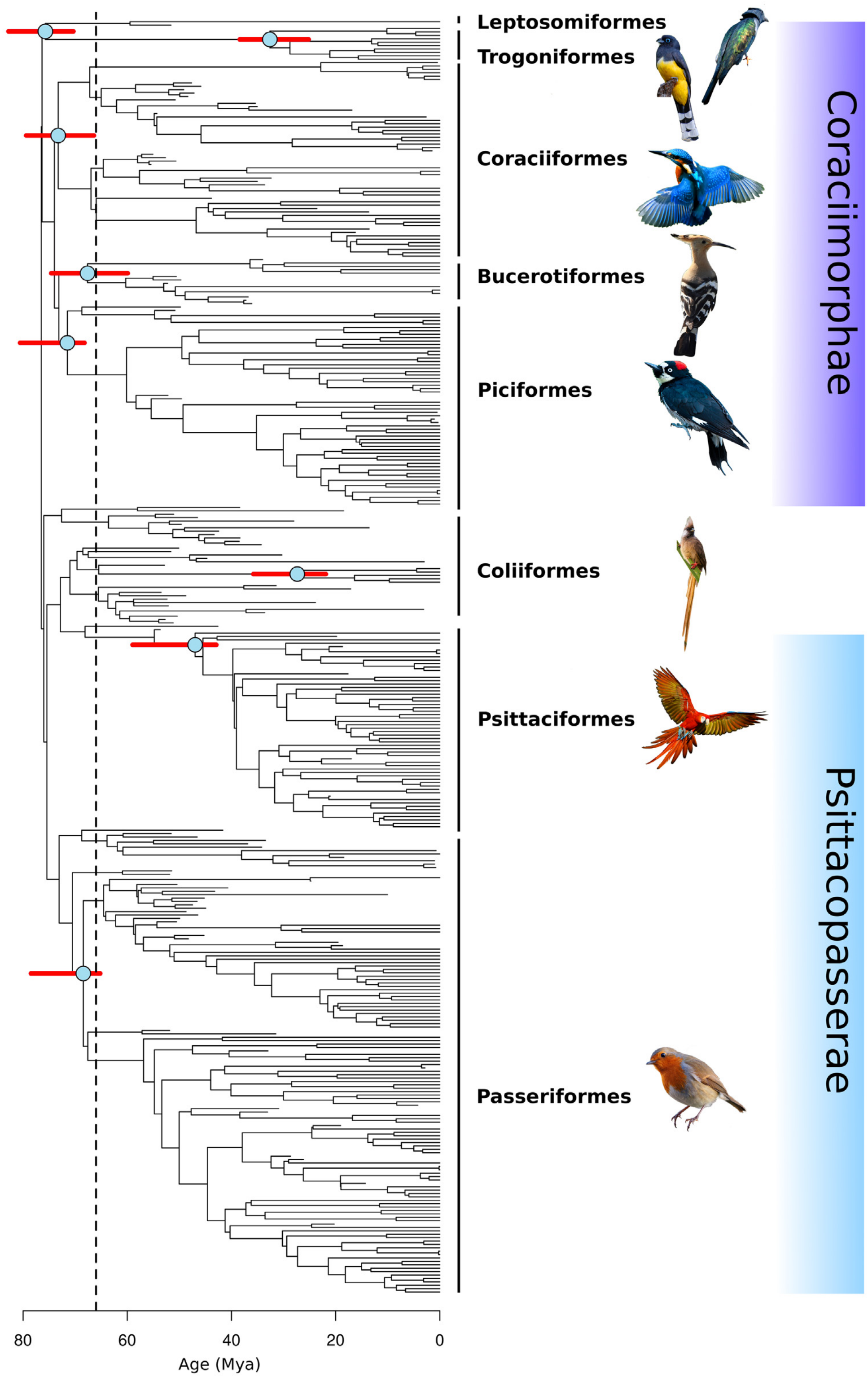

Fig. 3. A randomly sampled tree from the posterior distribution in the analysis of the combined data set. Red bars indicate the 95% confidence intervals for crown ages of each order. The passerine node highlighted is the oscine-suboscine split. Vertical dashed line indicates the K-Pg boundary. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

3.3. Fossil placement and taxonomy

Phylogenetic placement support for the extinct taxa was generally low, resulting in little overall support for published taxonomic classifications (Table S5). Only 37 out of 122 extinct taxa received majority support for placement within the same taxonomic order as in the original publications. Moreover, only 20 taxa had a placement support score of 0.70 or higher, indicating a broadly uncertain placement confidence with the available data. A few species had very high support for placement in a taxonomic order different than the originally

Table 1
Posterior crown ages (Mya) for Telluraves, each of the eight orders covered in this study, and for oscine and suboscine passerines. Ages are from 5000 samples drawn from the posterior tree distribution.

Order	Mean	95% HPD
Bucerotiformes	67.5	(59.9, 74.6)
Coliiformes	28.1	(21.9, 35.8)
Coraciiformes	73.6	(66.4, 79.4)
Leptosomiformes	76.6	(70.3, 82.8)
Passeriformes	71.8	(65.2, 78.5)
Oscines + Suboscines	69.0	(61.5, 75.4)
Piciformes	74.6	(68.2, 80.6)
Psittaciformes	50.9	(42.9, 59.0)
Trogoniformes	31.9	(25.2, 38.4)
(Telluraves)	76.6	(69.9, 82.8)

Table 2
Relative rate parameters for each morphological partition estimated by MrBayes. *n* is the number of characters present in the morphological matrix from each partition. Examples of characters from the 'other' partition include plumage, eggshell pattern and incubation behaviour.

Partition	*n*	Mean	Variance	Lower	Upper	Median
Skull	156	3.01	0.14	2.34	3.75	2.97
Wings	189	1.80	0.06	1.37	2.27	1.78
Body	115	1.21	0.03	0.91	1.55	1.19
Legs	307	1.36	0.01	1.16	1.57	1.35
Other	15	10.81	474.51	0.21	47.85	3.66

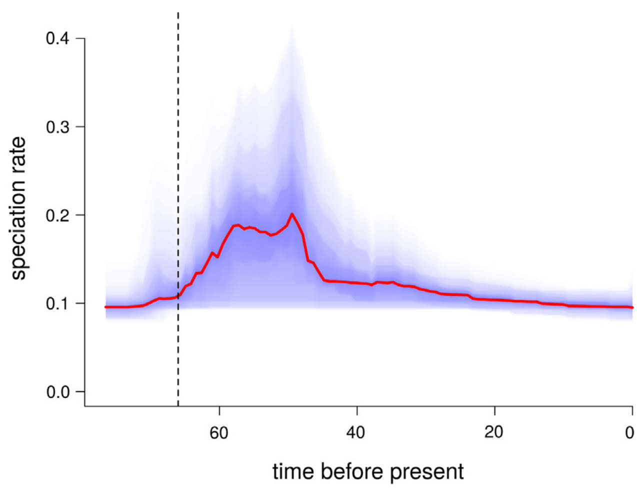

Fig. 4. Changes in estimated speciation rates though time. Vertical dashed line denotes the K-Pg boundary. Rates were estimated using BAMM v2.6 (Mitchell et al., 2018).

described. For example, *Sylphornis bretouensis*, thought to be a stem Piciform (Mayr, 2004, 2014), instead joined with Bucerotiformes with a 0.70 posterior probability. Four species show highest affiliation with the outgroup (nightjars & allies), outside of Telluraves, but support for these relationships is weak (*Chascacocolius cacicirostris*, placement support 0.43, *Eobucco brodkorbi*, 0.44, *Primocolius minor*, 0.45, & *Selmes MNH.Q.O.596*, 0.44). Topological support for the reconstruction was low (Fig. S10), likely a consequence of low placement support for extinct species.

The ANCOVA model explained a reasonably high proportion of the variance in placement support values, with an $r^2 = 0.51$ ($p < 0.0001$, Table 3). Three variables, the number of scored characters, diversity of body regions in scored characters and the number of characters with high phylogenetic signal, which we treated as independent, were each identified as significant predictors of placement support. The effect

Table 3
Mean and standard error of the effect sizes from ANCOVA of the placement support of extinct species. Below, we show the number of extinct species from the study (n), estimated parameters *k*, residual standard deviation (res. σ), r^2 is the variance explained by the model.

Coefficients	Estimate	Std. Error
Intercept	0.236	0.172
Number of scored characters	0.002*	0.001
Minimum age of species	−0.001	0.001
Diversity of scored body regions	0.224*	0.069
Super order (Coraciimorphae)	0.070	0.171
Super order (Psittacopasserae)	0.116	0.169
Number of characters with high phylogenetic signal	0.005*	0.001
n = 122		
$k = 7$		
Res. $\sigma = 0.17$		
$r^2 = 0.51$		

* Denotes effects estimated to be significant at the 0.05 level.

sizes for both number of characters and number of characters with high phylogenetic signal were, however, extremely low (0.002 and 0.005 respectively). On the other hand, the diversity of scored body regions had a large estimated effect size (0.224), suggesting that species whose scored characters were distributed across all of the major body regions were also commonly placed on the phylogeny with high confidence.

3.4. Phylogenetic signal

We estimated Pagel's λ for 113 morphological characters in the data matrix. Most of these characters belong to 'skull' ($n = 33$), with the remainder from the 'body' ($n = 23$), 'legs' ($n = 31$), 'wings' ($n = 25$), and 'other' ($n = 1$) regions respectively. For 109 characters, the estimated value was $\lambda \geqslant 0.80$, with a high overall mean ($\bar{\lambda} = 0.97$; Table S4). Despite some high λ estimates, 35 of the unconstrained models were not significantly different than a constrained model. Nevertheless, the high λ estimates suggest that the greatest morphological differences are between distantly related taxa.

4. Discussion

The date of origination of land birds continues to be hotly debated (Mitchell et al., 2015; Cracraft et al., 2015). Our total-evidence analyses estimate a crown age of approximately 76.6 Mya (69.9–82.8 Mya 95% HPD), suggesting some lineages were able to survive the K-Pg extinction. This result does not imply that Telluraves both originated and diversified in the Cretaceous, as suggested historically by neontological studies (Hedges et al., 1996; Cooper and Penny, 1997; Cracraft, 2001; van Tuinen and Hedges, 2001; Ericson et al., 2002, 2014; Barker et al., 2002; Jetz et al., 2012; Haddrath and Baker, 2012; Lee et al., 2014), with speciation rate estimates suggesting the group radiated immediately following the K-Pg boundary (Fig. 4). A Cretaceous crown age for Telluraves was not recovered by more recent large-scale neontological studies, however, and it has been noted that tip-dating methods consistently recover older divergence estimates compared with node-dating methods (Ronquist et al., 2012a; Slater, 2013; Arcila et al., 2015; O'Reilly et al., 2015). We find that the extent of the discord between tip-dating and node-dating results may be due to the number of extinct taxa sampled (and by proxy the amount of morphological data incorporated); such that including fewer extinct species resulted in younger age estimates. This may be because accounting for observed diversity changes the estimated rates of evolution (Wagner, 2000). An advantage of tip-dating is that a greater degree of taxonomic and morphological variation can be accommodated. Here, we have incorporated significantly more taxa than the most recent molecular studies, particularly Eocene taxa. For example, Prum et al. (2015) used the species *Sandcoleus copiosus* to calibrate the stem of mousebirds (Coliiformes), whereas here we have included 23 taxa (Table S6). Incorporating this diversity may be important to understanding the evolution of the group. Each dating method currently in use has a number of problems, and we suggest that these results highlight the need for continued discussion about reconciliation between the relative strengths of approaches, as well as best practices.

Our inferences of topology and node ages may be negatively affected by the relatively sparse morphological data matrix. Completeness of data matrices is an inherent problem in paleontological analyses, particularly those with broad taxonomic sampling (Fig. S3). Missing data can affect many different aspects of phylogenetic reconstruction (Simmons, 2012, 2014; Roure et al., 2013), including topology (Sansom and Wills, 2013; Sansom, 2014), branch length estimation (O'Reilly et al., 2015), and result in a variety of artifacts, such as apparently reduced support for groups that ought to be well supported, instead (Thomson and Shaffer, 2010). Of particular concern for our study, branch lengths can be artificially altered by missing data (Darriba et al., 2016) leading to inaccurate parameter estimation in models of sequence evolution (Heath et al., 2008), which may also affect our age estimates (as well as those in other studies). There are a few contributing factors to low coverage in our morphological data matrix. First, the available neornithine fossil record has particularly poor preservation rates of material compared to other groups, with a considerable number of species described from limited material. Second, individual studies, which focus on specific taxa, only perform phylogenetic reconstructions using the material preserved for that focal group. These characters are not necessarily shared between studies, so concatenated matrices from different studies can diminish overlap and overall coverage. Finally, as our sampling strategy relies on broad taxonomic coverage, we included many extant members of Passeriformes and Psittaciformes that were not included in published morphological matrices, which reduced the overall completeness of our matrix. On the other hand, there is a clear trade-off, in the sense that inclusion of additional taxa can increase the reliability of reconstructions by breaking up long branches, leading to greater confidence in parameter and topological estimation (Heath et al., 2008; Ronquist et al., 2012a; Crawley and Hilu, 2012). Incorporating extinct taxa reduces the gap to between the oldest Eoence fossils and the root to a smaller, but still large, 20 million years. This discrepancy may be due to an absence of specimens from the Late Cretaceous, across the K-Pg boundary, as a result of regional fluctuations in the completeness of the avian fossil record (Chiappe, 1995; Brocklehurst et al., 2012; Lee et al., 2014). Alternatively, the age of the avian crown may have been previously overestimated, if rates of molecular evolution were faster following the end-Cretaceous mass extinction due to differential survival of smaller bodied taxa (Berv and Field, 2018).

A striking result of this study is the recovery of Coliiformes as sister to the Psittaciformes (Fig. 2), rather than within the Coraciimorphae (Hackett et al., 2008; Jarvis et al., 2014; Prum et al., 2015). Rapid diversification at the base of the avian tree has long hindered the ability of systematists to resolve inter-ordinal relationships, as short internode distances provide little time for molecular substitutions, or morphological changes, to accumulate on branches. This problem has been greatly aided by the construction of larger genetic matrices which offer greater support to the relationships between extant taxa, notably by Jarvis et al. (2014, whole genomes of 48 species) and Prum et al. (2015, > 390 k bp from 198 species). The length of our genetic alignment (27 k bp) was sufficient to recover Coliiformes within the Coraciimorphae. Its estimated position in the tip-dating analysis is therefore probably due to the inclusion of the extinct taxa. Extinct taxa can erroneously alter inferred relationships if, for example, there is a preservation bias of plesiomorphic characters over synapomorphic characters (Sansom et al., 2010). Previous work by Mayr, 2013 has documented some of the morphological similarities between putative stem taxa and extant members of different orders. It is unclear if support for this relationship is an artifact of limited taxon and/or character sampling in the present study.

Our analyses also estimated a sister relationship between the Trogoniformes and monotypic Leptosomiformes with and without extinct taxa included. Typically, a pectinate branching pattern is recovered for the Coraciimorphae (Hackett et al., 2008; Jarvis et al., 2014; Prum et al., 2015). Estimating the position of Leptosomiformes has been historically challenging due to the extremely long branch leading to the single extant taxon (*Leptosomus discolor*). Greater confidence has been achieved recently with the construction of longer genetic alignments. For example, the node subtending *Leptosomus discolor* has a bootstrap support of 1 when analyzed by Jarvis et al. (2014) and Prum et al. (2015). The estimation of a sister relationship between Trogoniformes and Leptosomiformes here may be an artifact of our shorter genetic alignment. The discrepancies between the our topology and those from previous studies are unlikely to dramatically alter divergence estimates because the affected branches are inferred to have very short lengths (Fig. 3).

Our analysis recovered little confidence in the placement of extinct taxa. Only 20 of the 122 extinct taxa included support previously assigned taxonomy with a maximum support value of 0.70 or higher, with the remainder being of uncertain placement within the tree. This is possibly due to a combination of rapid diversification, rapid morphological differentiation during the early history of the group, paucity of character data, or inadequate models of morphological evolution. Typical solutions to resolving problematic taxonomic relationships in studies that use genetic data matrices have included the use of faster evolving genes and longer alignments (Philippe et al., 2011). Working with extinct taxa, however, means that there is a often a limit on the amount of data available. Our study results suggest that the some increase in data coverage could aid the placement confidence, but requires characters to be distributed across all body regions (Table 3). If additional material is not available, placing topological constraints on extinct taxa based on prior classifications is not a suitable solution. Doing so distorts branch length estimation which results in dramatically older divergence estimates (supplementary material).

Inclusion of a wide range of fossil Telluraves in our phylogenetic analyses appears to give some further insight into the evolution of the group. It is possible that a few lineages were able to survive the end-Cretaceous mass extinction before diversifying rapidly in the Paleogene. This period of lineage diversification was accompanied by rapid morphological evolution, giving rise to a wide range of forms by the mid-Eocene. The limitations of our data, however, leave room for expansion. These analyses will clearly benefit from the inevitably expanded acquisition of sequence data and tracking of improvements in implementation of models used for inference. The primary focus for future work should perhaps center on increasing the completeness of the morphological matrix. In particular, some of the easiest improvements could concern complete scoring of morphological data from extant taxa. High uncertainty will likely always accompany such analyses, but a combination of novel data sources and improved methods should result in greatly improved precision in the inferred timing of diversification of birds.

Authors' Contributions

N.C. helped conceive study, constructed the morphological alignment, performed phylogenetic reconstruction and drafted the manuscript; K.R. wrote script for and performed alignment of the genetic data; B.I. helped conceive the study and drafted the manuscript.

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Appendix A. Supplementary material

Supplementary data associated with this article can be found, in the online version, at https://doi.org/10.1016/j.ympev.2018.10.006.

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